Plato's Presence and Beyond: Co-Occurrence Networks in Ancient Greek and Latin Literature

de Graaf, Evelien
evelien.degraaf@kuleuven.be
KU Leuven, Belgium

Introduction

The study of (Social) Networks has steadily gained traction in the study of Graeco-Roman antiquity. For instance, the spread of religions throughout antiquity has been studied with approaches from network theory based on epigraphic attestations in certain places of certain deities of these religions (Collar 2013; 2022). Other studies in the field of prosopography, employ quantitative network approaches reconstructing Athenian kinship networks based on naming traditions (Karila-Cohen 2018), to facilitate the identification of individuals in cases of homonymy or to trace the spread of certain personal names in Graeco-Roman Egypt (Broux / Depauw 2015; Broux 2019). In the study of literary texts, the focus is on manually creating networks of people by reconstructing real-life interactions or relations with traditional close reading methods (Broekaert et al. 2020; Cline 2019; Guérin 2024; McClure 2020; Rollinger 2014). However, studies of automatically constructed networks are also emerging, exploring other types of connections in these literary texts (Beine 2024; Dörpinghaus 2022). Another closely related area of network research is the construction and analysis of Character Networks based on literary narratives, such as, for example, on Shakespeare (Ladegaard / Kristensen-McLachlan 2025) or contemporary Dutch literary fiction (Smeets 2021).

In this study, I model a co-occurrence network of individuals in Ancient Greek and Latin literature, similar to Dörpinghaus' (2022) approach for biblical narratives. This functions as a case study within the NIKAW (Networks of Ideas and Knowledge in the Ancient World) project that aims to study the flow of knowledge in Ancient Greek and Latin literature using Natural Language Processing (NLP) and Social Network Analysis (SNA). This case study serves as proof of concept for applicability of the co-occurrence network model, addressing concerns about the interpretability and meaningfulness of networks with edges based on textual proximity (Dörpinghaus 2022: 8; Guérin 2024: 8; Labatut / Bost 2019: 14). Plato, a prolific author with well-documented impact, serves as a model test case for assessing the usefulness of co-occurrence networks in understanding the reception of pivotal individuals over time. Additionally, his influence on both Christian and non-Christian authors allows for a comparative analysis of co-occurrences within these two distinct intellectual traditions. With this we build on previous research that focused on text reuse detection to understand connections between famous ancient authors (for Plato specifically, see the Digital Plato Project , Schubert et al. 2019). Our approach expands this by including all named individuals in texts, not just well-known authors, to address biases and map a broader network of influences.

Data

The corpus, detailed in Table 1, was constructed to meet several criteria: texts must contain the lemma 'Πλατων'/ 'Platon' more than five times, span a wide diachronic range, vary in text type (limited to prose) and authored by both Christians and non-Christians. Individuals in these texts are manually annotated and disambiguated. At the time of the presentation, the majority of the annotation will be finished.

Author	Work	Period	Christian work
Greek [GLAUx corpus]			
[Plato]	Epistulae	BCE	FALSE
Aristoteles	Metaphysica	BCE	FALSE
Strabo	Geographica	BCE	FALSE
Dionysius Halicarnassensis	De Demosthenis dictione	BCE	FALSE
Plutarchus	Quaestiones convivales	CE	FALSE
Claudius Aelianus	Varia historia	CE	FALSE
Clemens Alexandrinus	Stromata	CE	TRUE
Origenes	Contra Celsum	CE	TRUE
Diogenes Laertius	Vitae philosophorum	CE	FALSE
Galenus	De placitis Hippocratis et Platonis	CE	FALSE
Latin [LASLA / Clérice]			
Cicero	Tusculanae Disputationes	BCE	FALSE
Tertullian	De Anima	CE	TRUE
Seneca	Ad Lucilium Epistulae Morales	CE	FALSE
Lactantius	Divinarum Institutionum	CE	TRUE
Apuleius	Pro Se De Magia Liber	CE	FALSE

Table 1: Corpus for co-occurrence analysis.

This annotation facilitates the construction of a multi-modal graph as represented in the schema displayed in Figure 1. The nodes in this graph, represented by the circles, are Persons, Works, and Mentions and listed together with associated attributes. The edges in the original multi-modal graph encode the following relationships:

- A work contains a mention of a personal name: (:Work) – [:CONTAINS] -> (:Mention)
- The mention refers to an individual: (:Mention) – [:REFERS_TO] -> (:Person)
- A person wrote a work: (:Person) – [:WROTE] -> (:Work)

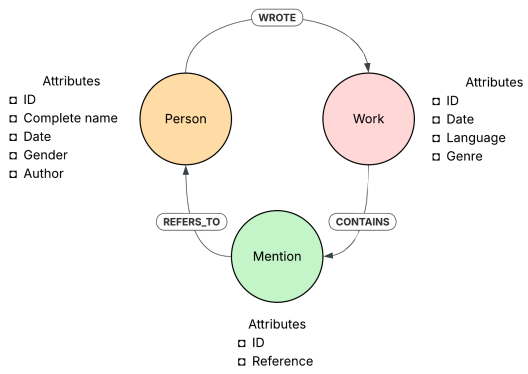

Figure 1: Schema of the multi-modal graph for mentions of people.

From this multi-modal graph we can project co-occurrence networks of persons at various levels of textual proximity, such as a range of words, sentences, paragraphs, etc. This projection would contain only Person nodes with a CO_OCCURS_WITH relation between two distinct individuals if they co-occur within a specific section of a text. This short paper will focus specifically on a preliminary description and analysis of this Person-Person co-occurrence projection. Edges are based on co-occurrence between individuals in sections of the text that mention Plato explicitly, thus limiting the network to individuals that occur in similar contexts as Plato.

Analysis

This analysis of co-occurrences aims to investigate the similarities and differences in co-occurrences within texts from pre-Christianity and the early Christian period to determine whether Plato was associated with different individuals in these distinct intellectual traditions. First, I will compare node rankings based on centrality measures in the pre-Christian and Early Christian networks, split into Christian and non-Christian works. In addition, community detection algorithms will be employed to identify clusters of co-occurring individuals to highlight differences between groups of mentions. Last, comparing additional network metrics such as diameter, density, and transitivity will further understanding of the differences between the networks. Excluding Plato from the networks will allow for the analysis to extend beyond Plato, demonstrating whether the observed similarities and differences apply to other individuals as well. The findings of this network analysis on co-occurrences of explicit mentions will be compared to general insights from the study of the

reception of Plato in antiquity to determine the extent to which co-occurrence networks reflect established scholarly understandings of influence (e.g. Tarrant et al. 2018; Schubert et al. 2019).

Conclusion

This case study will provide a new perspective on the use of explicit mentions of individuals in ancient literary texts. Comparing these findings with traditional reception studies on Plato aims to validate this approach and reveal where a new network approach can provide new insights. Methodologically, this study will outline the strengths and weaknesses of this approach, offering valuable insights for future uses of co-occurrence networks in historical texts.

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